6G Weekly Survey — May 11 - May 17, 2026

9 items across 3 topics, ranked by importance.

ISAC (sensing + communication)

Unconsented Sensing: A Sociotechnical Governance Framework for 6G ISAC

Anass Sedrati · arXiv cs.NI · 2026-05-08

★★★★■ Addresses a critical blind spot in current 6G standardization efforts; regulatory compliance will gate commercial ISAC deployment, making this a must-read for anyone designing real-world sensing-communication systems.

This position paper argues that 6G ISAC systems—turning cellular infrastructure into continuous sensing grids—face severe legal and ethical challenges that purely technical privacy measures (encryption, anonymization) cannot address. The author proposes a sociotechnical governance framework grounded in EU AI Act and GDPR compliance, emphasizing purpose-bound sensing activation, citizen transparency, and algorithmic accountability for downstream ML models. The work frames ISAC trustworthiness as a mandatory regulatory compliance problem, not just a cryptographic one.

Contribution: First systematic mapping of legal friction points between pervasive ISAC sensing and emerging digital rights legislation (EU AI Act, GDPR), proposing a concrete governance framework to ensure regulatory viability before infrastructure deployment.

Relevance to your work: Directly relevant—as ISAC research progresses toward prototype systems, understanding regulatory constraints on sensing activation, data use, and ML model deployment will shape feasible system architectures and data flows you design.

Holographic Surface Enabled Integrated Sensing and Communications

Haobo Zhang, Shuhao Zeng, Xinyuan Hu, Shupei Zhang, et al. · arXiv eess.SP · 2026-05-09

★★★★■ Strong architectural contribution addressing critical scalability and power consumption barriers in ISAC systems with experimental validation, though commercial viability of RHS technology at scale remains to be proven.

This paper presents holographic integrated sensing and communications (HISAC) using reconfigurable holographic surfaces (RHSs) as a cost- and energy-efficient alternative to massive MIMO phased arrays for 6G ISAC. RHSs employ leaky-wave antenna principles to enable ultra-massive MIMO with simplified hardware (no complex phase shifters or feeding networks), while holographic beamforming provides flexible joint communication-sensing beam synthesis under practical leakage power constraints. The work provides optimization frameworks, implementation details, and experimental validation for bidirectional ISAC enhancement (sensing-assisted communication and communication-assisted sensing).

Contribution: Introduces RHS-based holographic beamforming as a hardware-efficient ultra-massive MIMO realization for ISAC, addressing the unique leakage power constraint inherent to leaky-wave antennas while demonstrating practical system implementations.

Relevance to your work: Directly relevant to your ISAC research interests and AI-native air interface work, as RHS hardware constraints create novel optimization problems suitable for learned beamforming and potentially simplified neural receiver designs for holographic apertures.

Sensing-Aided Secure Multicast in Two-Level Rotatable Antenna-Enabled ISAC Systems: Modeling and Optimization

Zequan Wang, Liang Yin, Hao Xu, Yunan Sun, et al. · arXiv cs.IT · 2026-05-09

★★★■■ Solid theoretical contribution to ISAC physical-layer security with novel hardware architecture, though the practical feasibility of real-time mechanical antenna rotation for multicast scenarios remains unclear.

This paper proposes a two-level rotatable antenna (RA) architecture for secure multicast in integrated sensing and communication (ISAC) systems, where array-level and element-wise antenna rotations are jointly optimized with analog beamforming to enhance legitimate user signals while suppressing eavesdropper leakage. The authors derive Cramér-Rao bounds to quantify imperfect eavesdropper direction sensing, construct probabilistic angular uncertainty regions, and solve a max-min secrecy-rate optimization problem using manifold optimization and projected-gradient methods. Simulations demonstrate that the two rotation levels play complementary roles in simultaneously beamforming toward legitimate users and creating nulls over eavesdropper uncertainty regions.

Contribution: First application of two-level rotatable antennas to sensing-aided secure multicast, with CRB-aware optimization that accounts for imperfect eavesdropper angular estimation and probabilistic uncertainty regions.

Relevance to your work: Directly relevant to your ISAC research interest, demonstrating how sensing can enable physical-layer security by exploiting spatial degrees of freedom, though the mechanical rotation aspect may be less practical than electronic alternatives in 6G systems.

6G Physical Layer

Recurrent Transformer-Based Near- and Far-Field THz Wideband Channel Estimation for UM-MIMO

Dmitry Artemasov, Alexander Shmatok, Kirill Andreev, Alexey Frolov, et al. · arXiv cs.IT · 2026-05-12

★★★★■ Addresses a critical 6G challenge (hybrid near/far-field channel estimation for sub-THz UM-MIMO) with a practical deep learning architecture demonstrating strong performance gains and desirable generalization properties relevant to 3GPP's beyond-52.6 GHz studies.

The paper addresses THz UM-MIMO channel estimation where users span both near-field and far-field regions due to large antenna apertures and high frequencies. A block recurrent transformer with state memory is proposed that trains once and iteratively applies to estimate channels under hybrid beamforming constraints with limited RF chains. The method generalizes across varying scatterer distances, path counts, and wideband operation, achieving 5 dB (narrowband) and 7.5 dB (wideband) NMSE improvement over state-of-the-art.

Contribution: First application of recurrent transformers with state memory to unified near/far-field THz channel estimation that generalizes across propagation environments and bandwidth regimes with single-model training.

Relevance to your work: Directly addresses your sub-THz waveform and AI-native PHY interests by combining learned channel estimation with the emerging near-field problem in massive antenna arrays, using transformers for iterative inference under hardware constraints.

Stepped Frequency Division Multiplexing: A Jump-Free Continuous-Time AFDM Waveform

Yewen Cao, Yulin Shao · arXiv cs.IT · 2026-05-12

★★★★■ Addresses a concrete implementation barrier for AFDM deployment in 6G with rigorous analysis and a drop-in fix that preserves the IDAFT structure, likely relevant to standardization efforts targeting doubly-selective channels.

AFDM waveforms suffer from high out-of-band emission due to frequency wrapping that creates envelope discontinuities between sampling instants. The authors propose Stepped Frequency Division Multiplexing (SFDM), which holds instantaneous frequency constant at the chirp midpoint within each interval while accumulating phase continuously, eliminating jumps without changing IDAFT outputs. Simulation confirms OOBE reduction and improved robustness to fractional delays at the receiver.

Contribution: First analytical characterization of OOBE mechanism in continuous-time AFDM and a provably unique sample-preserving solution that removes internal discontinuities while maintaining full compatibility with standard AFDM processing.

Relevance to your work: Directly relevant: you track 6G waveform candidates including AFDM, and practical spectral compliance plus receiver robustness are critical for moving learned-PHY and AFDM prototypes toward real systems.

Look Once, Beam Twice: Camera-Primed Real-Time Double-Directional mmWave Beam Management for Vehicular Connectivity

Avhishek Biswas, Apala Pramanik, Eylem Ekici, Mehmet C. Vuran · arXiv cs.NI · 2026-05-06

★★★★■ Strong practical contribution with multi-domain validation (testbeds, datasets, vehicular experiments) addressing a critical 6G V2X beam management challenge, though camera dependence may limit adoption in all deployment scenarios.

This paper introduces VIBE, a camera-aided mmWave beam management system for vehicular V2X that combines ML-based initialization with model-based refinement and closed-loop RF feedback to achieve fast double-directional beam alignment. The hybrid architecture uses camera observations to prune the beam search space, then applies lightweight refinement and tracking to adapt to mobility and channel dynamics. Real-world testbed experiments and public dataset evaluations show VIBE achieves 1.1-1.4% outage rates, outperforming both 5G NR hierarchical beamforming and pure end-to-end ML approaches.

Contribution: The novelty lies in the hybrid architecture that leverages camera sensing for coarse beam-space reduction while maintaining model-based closed-loop adaptation, demonstrating that this fusion generalizes better to unseen vehicular scenarios than pure ML beam predictors.

Relevance to your work: Directly relevant as an example of hybrid AI/model-based beam management for mmWave massive MIMO, illustrating how multimodal sensing (camera + RF) can outperform end-to-end learning for real-time physical layer control in mobile scenarios.

NTN / satellite

A Disaster-Aware Integrated TN-NTN System-Level Simulator for Resilient 6G Wireless Networks

Donglin Wang, Anjie Qiu, Qiuheng Zhou, Hans D. Schotten · arXiv cs.NI · 2026-05-07

★★★■■ Solid infrastructure simulation tool addressing resilience—a key 6G requirement—but primarily a modeling framework rather than algorithmic or protocol innovation.

The authors develop a system-level simulator for integrated terrestrial-NTN 6G networks that models disaster scenarios with probabilistic base station failures and service migration to LEO satellites, HAPS, or UAVs. Following 3GPP Rel-17/18 principles, the simulator evaluates throughput, packet reception ratio, and latency under varying disaster severity and NTN provisioning levels. Results demonstrate the capacity-delay tradeoff of terrestrial operation versus the stability of NTN fallback, with hybrid TN-NTN achieving balanced resilience.

Contribution: First lightweight system-level simulator explicitly modeling disaster-induced TN failures with standardized NTN fallback, enabling quantitative resilience analysis for 6G integrated networks.

Relevance to your work: Provides a testbed for evaluating NTN integration strategies relevant to your work on non-terrestrial networks, though primarily focused on network-layer resilience rather than physical-layer AI or air interface design.

Comparative Analysis of Direct-to-Cell (D2C) and 3GPP Non-Terrestrial Networks (NTN) for Global Connectivity

Donglin Wang, Anjie Qiu, Qiuheng Zhou, Hans D. Schotten · arXiv cs.NI · 2026-05-07

★★★■■ Solid synthesis of two competing NTN paradigms with practical architectural recommendations, though primarily comparative analysis rather than novel technical contribution.

This paper compares two satellite connectivity paradigms: Direct-to-Cell (D2C) systems like Starlink that work with unmodified handsets for emergency coverage, versus 3GPP NTN standardized approaches designed for high-throughput broadband and deep 5G/6G integration. The analysis covers standardization, architecture, PHY innovations, and security across both approaches. Results indicate NTN provides superior performance and scalability for 6G, while a hybrid architecture combining terrestrial 5G, NTN broadband, and D2C fallback is proposed for safety-critical applications like autonomous vehicles.

Contribution: First comprehensive technical comparison of market-driven D2C versus standards-based 3GPP NTN architectures, proposing a tri-link hybrid model for 6G use cases requiring both broadband performance and emergency resilience.

Relevance to your work: Directly relevant to your NTN research interests: provides systems-level perspective on competing satellite architectures and their integration pathways for 6G, informing design choices for AI-native air interfaces in hybrid terrestrial-NTN scenarios.

Queue-Aware and Resilient Routing in LEO Satellite Networks Using Multi-Agent Reinforcement Learning

Mudassar Liaq, Mahyar Tajeri, Peng Hu · arXiv cs.NI · 2026-05-06

★★★■■ Solid incremental work addressing a real operational challenge in LEO constellations with a well-motivated multi-agent formulation, though the approach builds on established DRL techniques without fundamental algorithmic novelty.

This paper addresses routing in LEO satellite networks under dynamic topologies and time-varying traffic by proposing a multi-agent deep reinforcement learning framework where each satellite acts as an independent routing agent. The approach incorporates queue dynamics, background traffic, and link resilience into latency-aware routing decisions. Evaluations show ~50% overhead reduction versus Dijkstra while maintaining competitive latency and better scalability under high traffic loads.

Contribution: A distributed MA-DRL routing framework that explicitly models queue backlogs and resilience metrics for LEO networks, achieving practical overhead reduction compared to centralized shortest-path algorithms.

Relevance to your work: Directly relevant as LEO/NTN is within your scope; the multi-agent learning paradigm and queue-aware design could inform cross-layer optimization in 6G non-terrestrial access networks where satellite-ground integration requires adaptive resource management.